



Advanced Web Development

Members:

- Simbaña Michael
- Tufiño Erick
- Villagómez Doménica

NRC: 27819

Medical Appointment Web System

1. PROBLEM STATEMENT

A modern clinic needs an integrated web system that allows patients to schedule, manage, and track their medical appointments efficiently. The current process relies on phone calls and manual record-keeping, leading to appointment overlaps, loss of information, and delays in patient service.

- Overlapping or missed appointments
- Service delays and operational inefficiency
- Low patient satisfaction
- Difficulty managing resources and schedules
- No centralized patient health records

Both patient satisfaction and clinic operational efficiency are severely affected, highlighting the need for a digital, centralized solution.

2. PROPOSED SOLUTION

An integrated web-based appointment management system that provides:

For Patients:

- Book, modify, and view appointments online
- Receive automatic reminders (24h and 1h before appointment)
- Access their medical history (EHR)

- Complete satisfaction surveys
- Manage multiple appointments

For Doctors:

- Manage and visualize their consultation schedules
- Access consolidated calendar with all appointments
- View patient Electronic Health Records (EHR)
- Update appointment status and add diagnostic notes
- Optimize consultation time

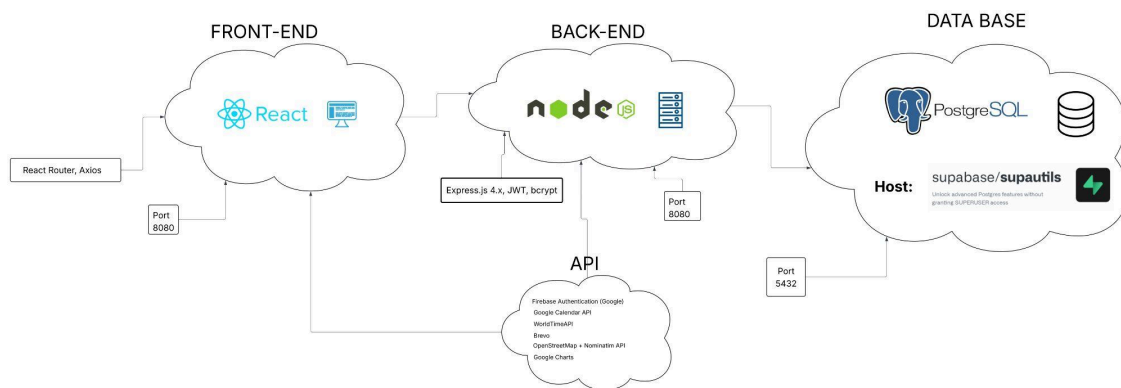
For Administrators:

- Manage users (patients, doctors, administrators)
- Manage physical resources (consultation rooms, equipment)
- Monitor system performance with key metrics
- Generate performance reports and billing
- Access audit logs for compliance

3. SYSTEM ARCHITECTURE

3.1 Architecture Overview

The Medical Appointment Web System follows a three-tier architecture pattern:



3.2 Frontend Architecture (React)

Component Structure:

- Common Components: Navigation, Footer, Loading Spinner
- Patient Components: Dashboard, Appointment Search, Booking, List, Survey
- Doctor Components: Dashboard, Schedule Config, Calendar, EHR Access, Notes

- Admin Components: Dashboard, User Management, Resource Management, Reports, Billing, Audit Log

Technologies:

- React 18+ with Context API for state management
- React Router v6 for client-side routing
- Axios for HTTP requests
- Google Charts for data visualization

3.3 Database Layer (PostgreSQL)

Tables:

- `users` - Base user information with roles
- `patients` - Patient-specific data (insurance, contact info)
- `doctors` - Doctor information (specialty, license)
- `administrators` - Admin permissions
- `appointments` - Main appointment records with status
- `medical\_records` - Patient EHR data
- `consultation\_rooms` - Physical resource management
- `doctor\_schedules` - Working hours and availability
- `billings` - Invoice and payment records
- `audit\_logs` - Complete action history for compliance
- `satisfaction\_surveys` - Patient feedback and ratings
- `reminders` - Notification tracking

4. FUNCTIONAL REQUIREMENTS (RF)

4.1 Patient Module (RF-P)

ID	Requirement	Description	Priority
RF-P.1	Secure Registration	Patient can register with basic info and email verification	High
RF-P.2	Authentication	Secure login with credentials or Google SSO (Firebase)	High

RF-P.3	Search Appointments	Search available appointments by doctor, specialty, or date	High
RF-P.4	Dynamic Slot Calculation	System automatically calculates available time slots	High
RF-P.5	Book Appointment	Reserve appointment with confirmation	High
RF-P.6	Modify/Cancel Appointment	Change or cancel scheduled appointments	High
RF-P.7	Automatic Reminders	Receive email/SMS reminders 24h and 1h before appointment	Medium
RF-P.8	Satisfaction Survey	Complete post-appointment feedback survey (1-5 rating)	Medium

4.2 Doctor Module (RF-D)

ID	Requirement	Description	Priority
RF-D.1	Configure Schedule	Set working hours, breaks, and exceptions	High

RF-D.2	Consolidated Calendar	View all appointments in single dashboard	High
RF-D.3	Access Patient EHR	View patient medical history and records	High
RF-D.4	Update Appointment Status	Change status (Scheduled→Confirmed→In Progress→Completed)	High
RF-D.5	Add Diagnostic Notes	Document consultation findings and treatment plan	High

4.3 Administrator Module (RF-A)

ID	Requirement	Description	Priority
RF-A.1	User Management	Create, read, update, delete users with role assignment	High
RF-A.2	Resource Management	Manage consultation rooms and equipment	High
RF-A.3	Performance Reports	Generate reports with key metrics (TC, AR, ACT, CR, PSI, OEI)	Medium
RF-A.4	Billing Generation	Create dynamic invoices based on consultations	Medium

RF-A.5	Audit Log Access	View complete history of system actions for compliance	Medium
--------	------------------	--	--------

5. NON-FUNCTIONAL REQUIREMENTS (RNF)

5.1 Performance (RNF-P)

ID	Requirement	Target
RNF-P.1	Search Response Time	≤ 3 seconds for appointment search
RNF-P.2	Concurrent Users	Support for 100+ simultaneous users
RNF-P.3	System Availability	99.5% uptime guarantee

5.2 Security (RNF-S)

ID	Requirement	Implementation
RNF-S.1	User Authentication	JWT tokens with secure session management
RNF-S.2	Data Encryption	HTTPS/TLS for transmission, encryption at rest

RNF-S.3	Authorization	Role-Based Access Control (RBAC) at all levels
---------	---------------	--

5.3 Quality (RNF-C)

ID	Requirement	Target
RNF-C.1	Usability	Intuitive interface for all user types
RNF-C.2	Maintainability	Clean, documented code following best practices
RNF-C.3	Portability	Compatible with Chrome, Firefox, Safari, Edge

6. SYSTEM ENTITIES

6.1 Main Entities and Relationships

Appointment Lifecycle:

States: Scheduled → Confirmed → In Progress → Completed/Cancelled

Related Entities:

- Patient (who is being seen)
- Doctor (who is attending)
- Consultation Room (where appointment occurs)
- Medical Record (patient health info)
- Billing (invoice generated)
- Reminders (notifications sent)
- Satisfaction Survey (feedback after completion)

Key Entities:

1. User- Base class with id, name, email, password, role
2. Patient - Extends User (date_of_birth, address, insurance_plan)
3. Doctor - Extends User (specialty, license_number)
4. Appointment - Records consultation (datetime, status, duration)
 - Status: Scheduled, Confirmed, In Progress, Completed, Cancelled
5. MedicalRecord - Patient EHR (allergies, diagnoses, treatments, prescriptions)
6. ConsultationRoom - Physical resource (capacity, equipment)
7. DoctorSchedule - Working hours (day_of_week, start_time, end_time, exceptions)
8. Reminde* - Notifications (appointment_id, type, sent_status)
9. SatisfactionSurvey - Feedback (appointment_id, rating, comments)
10. Billing - Invoices (appointment_id, amount, insurance_discount, status)
11. AuditLog - Action tracking (user_id, action, table_name, timestamp)
12. Administrator - Admin user (permissions list)
13. Report - Performance metrics (type, period, metrics)

7. FEATURE BREAKDOWN (PRODUCT BACKLOG)

Feature 0: Core CRUD Operations

Scope:

- Admin panel for user CRUD operations
- Resource (consultation room) management
- Role-based user creation
- Administrative dashboard foundation

Requirements Covered:

- RF-A.1 (User Management - Create, Read, Update, Delete)
- RF-A.2 (Physical Resource Management)

Deliverables:

- Admin user management interface
- Consultation room management interface

- User CRUD endpoints (create doctor, patient, admin)
- Resource CRUD endpoints (create/manage rooms)
- Admin dashboard with statistics
- User and resource listing with filters
- Authorization enforcement

Feature 1: Doctor Schedule Configuration

Scope:

- Doctor schedule management system
- Working hours configuration
- Break and exception handling
- Calendar display for schedule visualization

Requirements Covered:A

- RF-D.1 (Doctor configure schedule/working hours)
- RF-P.4 (Foundation for dynamic slot calculation)

Deliverables:

- Doctor schedule configuration endpoints
- Schedule display/edit interface
- Break and exception management
- Calendar visualization component
- Schedule validation system
- Availability checking foundation

Feature 2: Dynamic Slot Calculation Algorithm

Scope:

- Core slot availability calculation
- Appointment conflict prevention
- Multi-factor slot computation
- Performance optimization for <3 second response

Requirements Covered:

- RF-P.4 (Dynamic calculation of available time slots) - PRIMARY
- RF-P.3 (Appointment search foundation)

Deliverables:

- Complete slot calculation algorithm
- Optimized database queries
- Slot availability endpoint
- Conflict prevention logic

- Performance tuning and caching
- Algorithm documentation

Feature 3: Appointment Search & Booking

Scope:

- Patient-facing appointment search interface
- Advanced filtering capabilities
- Booking reservation system
- Confirmation workflow
- Appointment list management

Requirements Covered:

- RF-P.3 (Search appointments by doctor/specialty/date)
- RF-P.4 (Dynamic slot utilization)
- RF-P.5 (Book appointment)

Deliverables:

- Appointment search API with filters
- Search results interface
- Booking confirmation flow
- Appointment list view
- Appointment details display
- Search optimization

Feature 4: Appointment Modification & Cancellation

Scope:

- Appointment status management
- Modification of scheduled appointments
- Cancellation with reason tracking
- Availability re-checking for modifications

Requirements Covered:

- RF-P.6 (Modify/Cancel appointment by patient)

Deliverables:

- Appointment modification endpoint
- Appointment cancellation endpoint
- Modification UI with new slot selection
- Cancellation confirmation dialog
- Status transition validation
- Cancellation reason tracking

Feature 5: Automated Reminder System

Scope:

- Automated notification system
- Email and SMS reminders
- Scheduled reminder delivery
- Reminder tracking and logging

Requirements Covered:

- RF-P.7 (Automatic reminders 24h and 1h before appointment)

Deliverables:

- Brevo API integration (email/SMS)
- Reminder scheduler service
- Email templates for reminders
- SMS templates for reminders
- Reminder tracking database
- Retry logic for failed sends

Feature 6: Doctor Schedule & Calendar View

Scope:

- Doctor appointment calendar
- Schedule visualization
- Consolidated view of all appointments
- Calendar navigation and filtering

Requirements Covered:

- RF-D.2 (Consolidated appointment calendar for doctor)

Deliverables:

- Doctor calendar component
- Calendar data API endpoint
- Date range filtering
- Appointment status visualization
- Doctor dashboard calendar integration

Feature 7: Patient Electronic Health Records (EHR) & Consultation Notes

Scope:

- Medical record system for patient data and history
- Doctor EHR read/write, with note/versioning and status workflow
- Access control, audit trail, HTTPS security

- Note-taking during appointments (auto-save/version), diagnosis, prescriptions

Requirements Covered:

- RF-D.3 (Doctor access to patient EHR)
- RF-D.5 (Add diagnostic notes)
- RF-D.4 (Update status during consultation)
- RNF-S.2 (HTTPS/TLS for EHR)
- RNF-S.3 (Authorization: doctor/patient only)

Deliverables:

- Medical records database table/fields
- EHR endpoints for CRUD
- Consultation notes endpoints (version/history)
- EHR viewer/editor UI for doctor
- Strict access controls (only assigned doctor)
- Change timestamping/audit
- Audit trail for all EHR/notes

Feature 8: Analytics, Surveys & Billing

Scope:

- Analytics dashboard: visualizations, trends, drill-down, export (CSV/PDF)
- Post-appointment survey: user feedback, PSI metric
- Invoicing/billing: dynamic calculation, payment status, invoice export
- Audit logging & regulatory compliance: complete audited log, UI for logs, retention

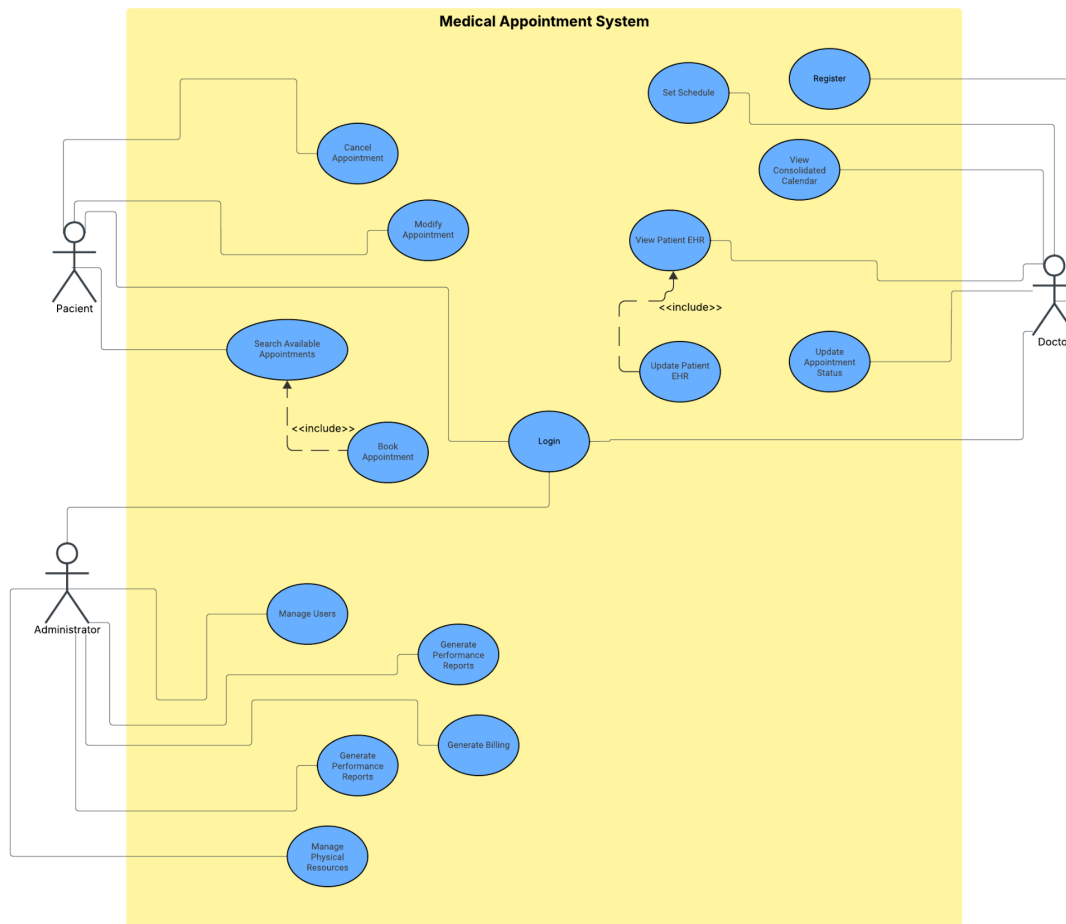
Requirements Covered:

- RF-A.3 (Reports with all metrics)
- RF-A.4 (Billing generation)
- RF-A.5 (Audit log access)
- RF-P.8 (Satisfaction survey)
- RNF-P.2 (100+ users/metrics)
- RNF-C.1 (Professional analytics display)
- RNF-C.2 (Maintainable metric/billing code)

Deliverables:

- Metrics calculation service, dashboards, API endpoints
- Feedback survey forms/results/tracking
- Billing engine, invoice export (PDF/CSV), payment tracking
- Audit log/report export UI
- Advanced analytics dashboard: charts, date filtering, drill-down
- Test suites (unit, integration, E2E), optimization, deployment documentation
- Security and performance validation
- User/developer guides, deployment runbook

9. USE CASES



9.1 Patient Use Cases

UC1: Register - Patient creates new account with email verification

UC2: Login - Patient authenticates with credentials or Google SSO

UC3: Search Available Appointments - Patient finds appointments by doctor/specialty/date

UC4: Book Appointment - Patient reserves available time slot

UC5: Modify Appointment - Patient changes scheduled appointment

UC6: Cancel Appointment - Patient cancels scheduled appointment

UC7: Send Appointment Reminder - System automatically sends email/SMS notifications

UC8: Complete Satisfaction Survey - Patient rates appointment experience

UC9: View Appointment History - Patient sees all past and future appointments

UC10: View Patient EHR - Patient accesses their medical records

9.2 Doctor Use Cases

UC11: Set Schedule - Doctor configures working hours and breaks

UC12: View Consolidated Calendar - Doctor sees all appointments in one view

UC13: View Patient EHR - Doctor accesses patient medical history

UC14: Update Appointment Status - Doctor changes appointment status during consultation

UC15: Update Patient EHR - Doctor adds consultation findings and prescriptions

9.3 Administrator Use Cases

UC16: Manage Users - Administrator creates/updates/deletes user accounts

UC17: Manage Physical Resources - Administrator manages consultation rooms and equipment

UC18: Generate Performance Reports - Administrator generates metric reports

UC19: Generate Billing - System generates invoices from completed appointments

UC20: View Audit Log - Administrator reviews system action history

10. EXTERNAL API INTEGRATIONS

10.1 Authentication & Authorization

Firebase Authentication

- Provider: Firebase (Google)
- Purpose: Secure Single Sign-On (SSO) for all users
- Usage: Alternative login method using Google accounts
- Implementation: Firebase SDK on frontend, token verification on backend

10.2 Calendar & Time Management

Google Calendar API

- Provider: Google Cloud
- Purpose: Synchronize doctor schedules with personal Google Calendars
- Usage: When doctor updates schedule in system, changes reflect in their Google Calendar
- Requirements: OAuth2 authentication, Calendar API scope

WorldTimeAPI

- Provider: Open-source time API
- Purpose: Handle timezone conversions accurately
- Usage: Display appointment times in patient's local timezone
- Endpoint: `https://worldtimeapi.org/api/timezone/{timezone}`

10.3 Communication Services

Brevo (formerly Sendinblue)

- Provider: Brevo
- Purpose: Email and SMS notifications
- Usage Scenarios:
 - Appointment confirmations (email)
 - Appointment reminders 24h before (email/SMS)
 - Appointment reminders 1h before (SMS)
 - Password reset emails
 - Survey links and notifications
- Integration: REST API with API key authentication

10.4 Reporting & Visualization

Google Charts

- Provider: Google
- Purpose: Data visualization for performance reports
- Usage:
 - Render line charts for metric trends
 - Display bar charts for doctor/specialty comparisons
 - Show pie charts for resource utilization
- Interactive, exportable charts